Current Transformers

Theory supplement to the CT experiment: errors, burden, saturation, the Thomson bridge and the two test methods

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1. Purpose

This handout gives the background for the current transformer (CT) experiment. The lab instructions refer to the lecture notes for several definitions; this document collects them in one place, in the order you meet them in the lab sheet (CT lab instructions). It covers:

- the equivalent circuit and why a CT has errors at all;
- the current error, phase displacement and composite error of IEC 61869-2, and the class limits you compare against;
- the burden, saturation, the instrument security factor FS and the secondary limiting e.m.f.;
- the Thomson bridge used to measure the winding resistance;
- the logic of the direct and the indirect test methods.

2. What a CT does

A CT reproduces the primary current at a reduced, standard secondary level (typically 1 A or 5 A), so that meters and protection relays can be connected safely and with a standard input. Its primary winding is in series with the line; its secondary feeds the burden. Two families exist:

- **Measuring CTs** (classes such as 0.5 or 1) must be accurate around rated current and should saturate early during faults, so that connected meters are protected.
- **Protective CTs** (classes such as 5P or 10P) must stay accurate up to many times rated current, so that relays see the fault current correctly.

The terminals are marked P1 and P2 (primary, older marking K and L) and S1 and S2 (secondary, k and l). Current entering P1 leaves the secondary at S1, which fixes the polarity for phase measurements.

The lab CT has a transformation ratio of 1, so the primary and secondary currents are nominally equal. This is what allows the difference current $i_d = i_1 - i_2$ to be measured directly with an ammeter in the direct method (Section 8).

The CT used in the lab

The lab board carries three identical CTs, numbered 1 to 3. Each has large primary sockets K and L and small secondary sockets k and l. The rating plate (Figure 1) reads:

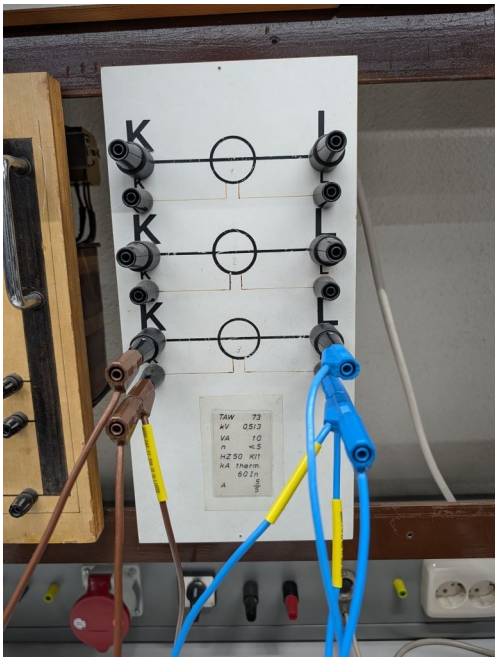


Figure 1. The CT board of the Protection lab with the rating plate. In this photo the leads are connected to CT 3.

Table 1. Rating plate of the lab CT and its meaning

Marking	Value	Meaning
TAW	73	Type designation
kV	0.5/3	Highest voltage for equipment 0.5 kV; insulation test voltage 3 kV
A	5/5	Rated primary and secondary current 5 A: ratio $K_n = 1$
VA	10	Rated burden 10 VA, i.e. $Z_{BN} = 10\ \text{VA} / (5\ \text{A})^2 = 0.4\ \Omega$
Kl	1	Accuracy class 1 (measuring CT)
n	< 5	Overcurrent factor in the older German notation, the equivalent of FS 5
Hz	50	Rated frequency
kA therm.	60 I_n	Rated short-time thermal current $60 \times I_n = 300$ A

Safety. Never open the secondary circuit while primary current flows. The CT then tries to drive its current into an infinite impedance, the core saturates and dangerous voltages can appear across the open terminals. Short the secondary before changing the burden.

3. Equivalent circuit and the origin of the errors

Figure 2 shows the equivalent circuit referred to the secondary side. An ideal transformer with ratio K_n delivers the current I_1/K_n. Part of it, the exciting current I_0, is needed to magnetise the core and flows through the magnetising branch L_m. Only the rest, I_2, reaches the burden Z_B.

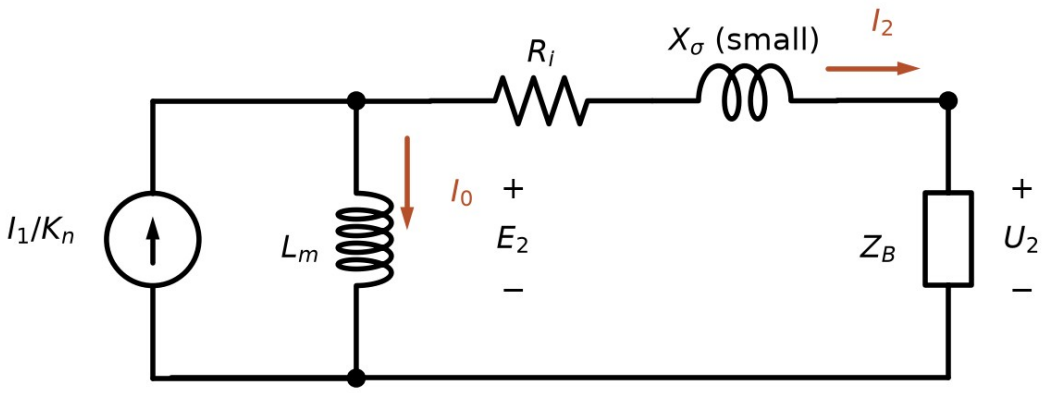


Figure 2. Equivalent circuit of an inductive CT referred to the secondary. R_i: secondary winding resistance; X_σ: leakage reactance; L_m: magnetising inductance; Z_B: burden.

The voltage the core has to support is the internal e.m.f.

$$E_2 = I_2 \cdot (R_i + jX_\sigma + Z_B)$$

For a ring-core CT with an evenly distributed secondary winding the leakage reactance is small, so the internal impedance is close to the winding resistance ($Z_i \approx R_i$), as the lab sheet assumes. This is why R_i is measured before the test.

The exciting current depends on E_2: a larger burden or a larger current requires a larger e.m.f., hence more flux and more exciting current. Every CT error comes from I_0, as the phasor diagram in Figure 3 shows: the flux Φ lags E_2 by 90°, I_0 is mostly in phase with Φ, and $I_1/K_n = I_2 + I_0$. As a result I_2 is slightly smaller than I_1/K_n (negative current error) and slightly leads it (positive phase displacement).

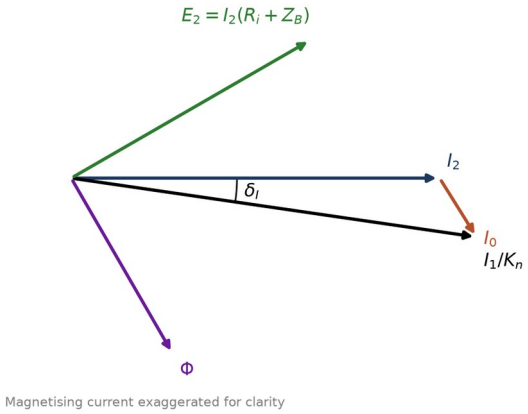


Figure 3. Phasor diagram of a CT. I_0 is drawn much larger than in a real CT so that the angles are visible.

4. Errors defined in IEC 61869-2

The lab sheet uses the three standard errors of IEC 61869-2. With the rated transformation ratio K_n:

Current (ratio) error. The difference in RMS magnitude, in per cent of the primary current:

$$\varepsilon_I = (K_n \cdot I_2 - I_1) / I_1 \cdot 100\ \%$$

This is the definition in the standard (EN 61869-2 preview). A negative value means the CT reads low.

Phase displacement. The angle between the primary and the secondary current phasors, $\delta_I = \arg(I_2) - \arg(I_1)$. With this sign convention δ_I is positive when the secondary current leads. It is normally given in minutes of arc (1° = 60′).

In the lab you read the time shift Δt between the two currents on the oscilloscope. At 50 Hz one period (20 ms) corresponds to 360° = 21 600′, so

$$\delta_I\,[\prime] = 360 \cdot 60 \cdot f \cdot \Delta t = 1.08 \cdot \Delta t\ [\mu s] \quad (\text{at } f = 50\ \text{Hz})$$

A time shift of 10 µs is therefore only about 11 minutes. Read Δt at the zero crossings, with the time base as fast as possible, and use calibrated current clamps, because the clamps themselves can add a phase shift.

Composite error. The RMS value of the difference between the ideal and the actual secondary current over one period, in per cent of the primary RMS current:

$$\varepsilon_C = (100 / I_1) \cdot \sqrt{\left((1/T) \int_0^T (K_n \cdot i_2(t) - i_1(t))^2\, dt \right)}$$

Unlike ε_I and δ_I, the composite error also captures the waveform distortion caused by saturation, which is why it is the quantity that defines FS and the protective classes. For a ratio-1 CT the integrand is just the instantaneous difference current, so $\varepsilon_C = I_d / I_1$ with I_d read on a true-RMS ammeter.

Class limits you compare against

For measuring CTs the current error and phase displacement must stay within the limits of Table 2 for any burden between 25 % and 100 % of the rated burden (EpiSensor summary of IEC 61869-2). Task b) of the lab asks you to check class 1.

Table 2. Limits for a class 1 measuring CT (IEC 61869-2, Table 201)

Primary current in % of I_{1n}	5 %	20 %	100 %	120 %
Current error ±%	3.0	1.5	1.0	1.0
Phase displacement ±minutes	180	90	60	60

Sources for these values: meteringlab.com and a LEM datasheet tested to Table 201.

Table 3. Limits for protective CTs (IEC 61869-2, Table 205)

Class	Current error at I_{1n}	Phase displacement at I_{1n}	Composite error at the accuracy limit current
5P	±1 %	±60 minutes	5 %
10P	±3 %	not specified	10 %

Source: Arcteq summary of IEC 61869-2. A protective CT is written for example as 5P20: 5 % composite error at 20 times rated current with rated burden.

5. Burden

The burden Z_B is the total impedance connected to the secondary: relay or meter inputs, the cabling and any added resistor. It is stated either in ohms or as an apparent power at rated secondary current with a power factor, typically 0.8 inductive to 1. The two are related by

$$S_N = I_{2n}^2 \cdot Z_{BN}$$

Example: a rated burden of 10 VA corresponds to 10 Ω for a 1 A secondary but only 0.4 Ω for a 5 A secondary such as the lab CT. For 5 A CTs the cabling alone can therefore be a noticeable part of the burden, which is why the lab sheet asks you to measure the burden resistors including their cables.

In the lab the burden is a board with three adjustable resistors (Figure 4): Rü for overloading (Überlast), R_N for the rated burden (Nennbürde) and R_U for underloading (Unterlast). They are purely resistive, so the burden power factor is 1 and the resistive approximations below apply directly.

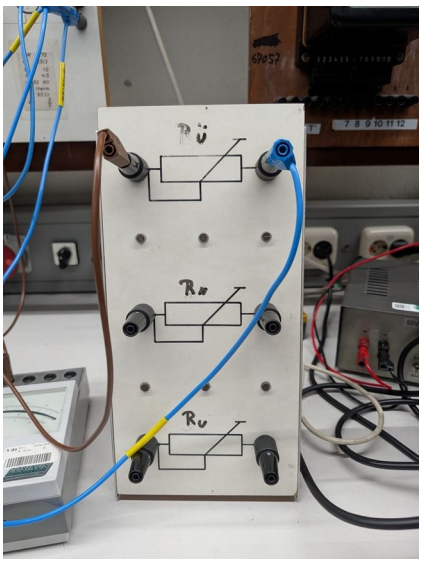


Figure 4. Burden board with the adjustable resistors Rü (overloading), R_N (rated) and R_U (underloading). The brown and blue leads connect the board to CT 3.

The burden matters because it sets E_2 for a given current (Section 3): a larger burden means more flux, more exciting current, larger errors and earlier saturation. In the lab you test the CT with the rated burden, a larger one (overloading) and a smaller one (underloading).

6. Saturation, FS and the limiting e.m.f.

The core of an inductive CT has a nonlinear magnetisation curve. Seen from the secondary terminals it appears as the excitation curve $U_2 = f(I_2)$ of Figure 5: in the linear region the exciting current is tiny; above the knee a small increase in voltage needs a large increase in exciting current. Once the e.m.f. required by the burden exceeds this region, the core saturates for part of each half cycle and the secondary current collapses and becomes strongly distorted.

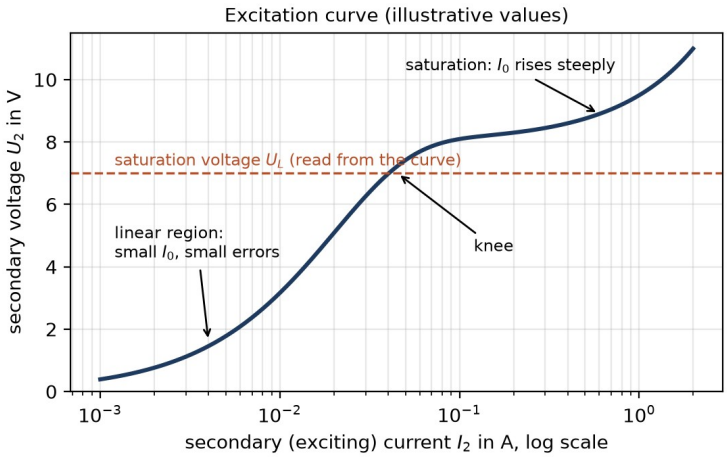


Figure 5. Excitation curve of a CT, measured from the secondary with the primary open (indirect method). Illustrative values.

Instrument security factor FS. For measuring CTs, FS is the ratio of the rated instrument limit primary current to the rated primary current. The rated instrument limit primary current is the smallest primary current at which the composite error reaches 10 %, with rated burden.

These definitions are from the standard (EN 61869-2 preview). A low FS (typical values 5 or 10) is a feature: the meter is protected because the CT saturates early during a fault.

Secondary limiting e.m.f. The product of FS, the rated secondary current and the vectorial sum of the rated burden and the secondary winding impedance:

$$E_{FS} = FS \cdot I_{2n} \cdot \sqrt{((R_i + R_B)^2 + X_B^2)}$$

where R_B and X_B are the resistive and reactive parts of the rated burden (IEC 61869-2 sample). For protective CTs the same idea is used with the accuracy limit factor instead of FS. This is the voltage U_H of the lab sheet, and the voltage at which you expect the knee in the indirect test.

FS at a different burden

E_{FS} is a property of the core. If the actual burden differs from the rated one, the core reaches the same e.m.f. at a different current, so the effective security factor changes. With a purely resistive approximation:

$$FS_x \approx FS \cdot (R_i + Z_{BN}) / (R_i + Z_B)$$

This is the formula you compare with in task g) (kVA Tools, after IEC 61869-2 §3.4.205). Underloading increases the effective FS and overloading reduces it (Figure 6). The standard itself warns that with a much lower burden the meter sees larger currents during a short circuit.

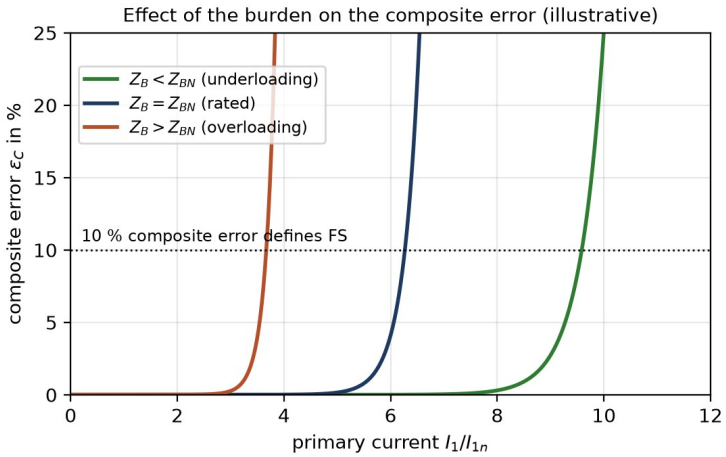


Figure 6. Composite error against primary current for three burdens. The current at which ε_C reaches 10 % gives FS for each burden. Illustrative curves, not the lab CT.

7. Measuring the winding resistance

Before the test you measure the secondary winding resistance R_i, which for a 5 A CT is typically well below 1 Ω. An ordinary ohmmeter or a Wheatstone bridge is not suitable at this level, because the resistance of the test leads and of the contacts is of the same order as the quantity being measured and is added to the result.

The Thomson bridge, also called the Kelvin double bridge, solves this with four-terminal (Kelvin) connections. Heavy current leads carry the measuring current through the reference resistor R_n and the unknown R_x; separate potential leads, which carry almost no current, pick up the voltages directly at the resistors (Wikipedia, Thomson-Brücke).

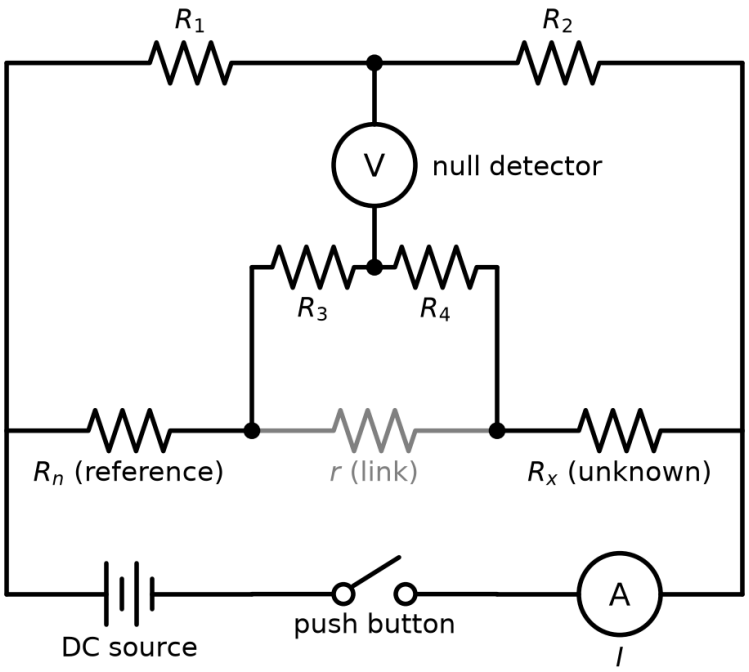


Figure 7. Principle of the Thomson (Kelvin double) bridge. R_1 and R_2 are the outer ratio arms, R_3 and R_4 the inner arms across the link r.

Principle: the Thomson bridge

The two pairs of ratio arms are adjusted together, so that their ratios are always equal. When the null detector shows zero, the bridge is balanced and

$$R_1 / R_2 = R_3 / R_4 \qquad \text{and} \qquad R_x = R_n \cdot R_2 / R_1$$

The inner arms R_3 and R_4 compensate the voltage drop on the link r between R_n and R_x, so r, the lead resistances and the contact resistances drop out of the result (Wikipedia, Thomson-Brücke, Elektroniktutor, TU Wien exercise). Thomson bridges cover roughly 1 Ω down to about 100 µΩ with good accuracy.

The instrument used in the lab: a four-wire milliohmmeter

Instead of an analogue bridge the lab uses a Gossen Metrawatt METRA HIT 27I (Figure 8), a handheld multimeter with a milliohm function and Kelvin (four-wire) connection (Gossen Metrawatt). It applies the same idea as the Thomson bridge in a simpler way: it drives a known DC current through the winding on one pair of leads and measures the voltage drop with a second pair, and $R_x = U / I$. Lead and contact resistances carry the current but are outside the voltage measurement, so they do not enter the result.



Figure 8. METRA HIT 27I milliohmmeter with its four-wire lead set. At each end one black current lead and one grey sense lead go to the same terminal of the device under test.

- **Terminals.** The current is fed through the ⊥ and V/Ω jacks; S− and S+ are the sense inputs for the four-wire connection.
- **Ranges.** The mΩ setting uses a DC measuring current of 200 mA (30 mΩ and 300 mΩ ranges) or 20 mA (3 Ω and 30 Ω ranges). The mΩ@1A setting uses 1 A pulses for 3 to 300 mΩ.
- **Inductive objects.** According to the manual, the DC mΩ method is suitable for resistances with inductances of up to 1 H, which is the relevant case for a CT winding.

Details from the operating manual. Connect the current and sense leads of each end to the same socket (k or l, or K or L), with the sense contacts as close to the winding as possible.

Use the meter setting given by the lab staff for the CT winding.

Practical notes for the CT winding

- Disconnect the CT from the test setup first, as the lab sheet requires.

- A CT winding is an inductance. Wait until the reading has settled; with the bridge, release the push button before disconnecting.
- The DC current can leave some remanent flux in the core. If the following AC results look odd at low current, demagnetise the core by slowly raising and lowering an AC current.
- Copper resistance rises by about 0.39 % per kelvin. Note the room temperature; R_i measured cold is slightly lower than during operation.
- Table 1 of the lab sheet asks for R_i seen from both sides. For a ratio-1 CT the two windings have similar but not necessarily equal resistance; use the secondary value in $E_2 = I_2 (R_i + Z_B)$.

8. Direct measurement (lab section 4.2)

The CT is supplied from a variable transformer T1 through a current-limiting resistor R_B, and the burden is connected to the secondary. Because the ratio is 1, an ammeter connected between the two windings carries the difference current $i_d = i_1 - i_2$ directly, which gives the composite error without a reference CT.

- **True-RMS instruments.** Near saturation the currents are not sinusoidal. Average-responding meters scaled for a sine wave then read incorrectly; only TRMS meters give the RMS values that the definitions need.
- **Phase.** Measured from the time shift between the two clamp signals (Section 4).
- **Current steps.** The measurement points 5 %, 20 %, 100 % and 120 % of I_{1n} match the points of Table 2, so you can check the class directly.

Repeating the test with the three burdens gives the curves of Figure 6 and, from the 10 % crossing, the FS for each burden.

9. Indirect measurement (lab section 4.3)

A direct test up to FS times rated current needs a large current source. The indirect method avoids this. The primary is left open and the secondary is supplied with a sinusoidal 50 Hz voltage through the high-current transformer T2. All the current that flows is then exciting current, and U_2 versus I_2 is the excitation curve (Figure 5).

The idea is that, in service, the core does not know whether its e.m.f. comes from a primary current or from an external source. At a given e.m.f. U_2, the CT in service would deliver the secondary current

$$I_2' = U_2 / (Z_B + R_i)$$

and the exciting current I_2 that you measured at that voltage is the current lost in the core. The composite error is then approximately

$$\varepsilon_C \approx I_2 / I_2'$$

These are the two equations of task j). Repeating the calculation for the three burdens gives FS for each from a single measured curve.

- **Expected U_H.** Calculate E_{FS} from Section 6 before the test and choose the tap of T2 (3 V, 6 V or 24 V) so that you can reach it. With the plate values and R_i of the order of 0.1 Ω, E_{FS} is above 12 V (Section 10), so only the 24 V tap reaches it.
- **Point density.** Take small steps near the knee; the curve changes fastest there.

- **Stop criterion.** Stop at U_H or when I_2 reaches about 20 % of the rated secondary current, as the lab sheet specifies.

Expect some difference between the two methods (task l). The indirect method neglects the leakage reactance and the waveform distortion of the secondary current, assumes that the burden behaves like the resistance used in the formula, and is affected by remanence and by the temperature of the winding.

10. Worked example with the lab CT

The rating plate values are those of the lab CT. The winding resistance and all measured values below are assumed, to show the steps; replace them with your own measurements.

- Plate: 5 A / 5 A, class 1, FS 5, rated burden 10 VA. Assumed $R_i = 0.1\ \Omega$.
- Rated burden: $Z_{BN} = 10\ \text{VA} / (5\ \text{A})^2 = 0.4\ \Omega$ (resistive).
- Limiting e.m.f.: $E_{FS} = 5 \cdot 5\ \text{A} \cdot (0.1 + 0.4)\ \Omega = 12.5\ \text{V}$. The 24 V tap of T2 is needed for the indirect test.
- Overloading with an assumed $Z_B = 0.8\ \Omega$: $FS_x \approx 5 \cdot 0.5 / 0.9 \approx 2.8$. Underloading with $Z_B = 0.2\ \Omega$: $FS_x \approx 5 \cdot 0.5 / 0.3 \approx 8.3$.
- Direct test at 100 % I_{1n}: $I_1 = 5.00$ A, $I_2 = 4.96$ A gives $\varepsilon_I = (4.96 - 5.00) / 5.00 = -0.8$ %, inside the ±1 % of class 1.
- Time shift $\Delta t = 40$ µs gives $\delta_I \approx 1.08 \cdot 40 = 43'$, inside the ±60′ of class 1.
- Indirect test with rated burden: at $U_2 = 3.0$ V the measured exciting current is 0.05 A. Then $I_2' = 3.0 / 0.5 = 6$ A and $\varepsilon_C \approx 0.05 / 6 \approx 0.8$ %.
- Milliohmmeter: a current of 200 mA and a sensed voltage of 20 mV give $R_i = 0.020\ \text{V} / 0.2\ \text{A} = 0.1\ \Omega$.

11. Link to the post-processing tasks

Task in the lab sheet	Section of this handout
a), b) ε_I, δ_I, ε_C at rated burden; class 1 check	4
c), d) overloading and underloading; $I_2 = f(I_1)$	5, 8
e), f), g) FS from $\varepsilon_C = f(I_1)$, U_H, FS_x for other burdens	6
h) $\varepsilon_C = f(U_2)$	3, 6
i) excitation curve and saturation voltage U_L	6, 9
j), k), l) indirect method and comparison	9

References

- IEC 61869-2:2012, Instrument transformers, Part 2: Additional requirements for current transformers. Sample pages: https://cdn.standards.iteh.ai/samples/16154/0ba92e2034f4419685fc8cd7ae2c2ec9/IEC-61869-2-2012.pdf
- EN 61869-2:2012, preview of definitions 3.4.201 to 3.4.206: https://standards.iteh.ai/catalog/standards/clc/425e3c62-aa8f-49f7-a01b-aa1b033647c0/en-61869-2-2012
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- Elektroniktutor, Die Widerstandsmessbrücke, Thomson-Messbrücke: https://www.elektroniktutor.de/analogtechnik/bruecke.html
- TU Wien ACIN, Messtechnik exercise 3, Thomson bridge: https://www.acin.tuwien.ac.at/file/teaching/bachelor/Messtechnik/376045_%C3%9Cbung3.pdf
- Lecture notes for the module. Available from the lab staff.